

Assignment 1

Your Name

Due: October 15, 2026

Problem 1: Triangle inequalityShow that $|x + y| \leq |x| + |y|$ for all $x, y \in \mathbb{R}$.

Solution. Adding $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$ gives

$$-(|x| + |y|) \leq x + y \leq |x| + |y|.$$

The claim follows from the definition of absolute value. □**Problem 2(a): A convergent sequence**Prove that $a_n = 1/n$ converges to 0.

Solution. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ with $N > 1/\varepsilon$. For every $n \geq N$, we have $|a_n - 0| = 1/n \leq 1/N < \varepsilon$. Hence $a_n \rightarrow 0$. □

Problem 2(b): A shifted sequenceDeduce that $b_n = 3 + 1/n$ converges to 3.

Solution. Since $|b_n - 3| = 1/n$, the same estimate as in Problem 2(a) proves the claim. □

Problem 3: An integralEvaluate $\int_0^1 x^2 \, dx$.

Solution. By the fundamental theorem of calculus,

$$\int_0^1 x^2 \, dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3}.$$

□